

MAYANK GAUR

[LinkedIn](#) [GitHub](#)

[✉ mayankgaur13004@gmail.com](mailto:mayankgaur13004@gmail.com)

[☎ +91-8840950313](tel:+91-8840950313)

[📍 Lucknow, Uttar Pradesh, India](#)

EDUCATION

- Dr. A.P.J. Abdul Kalam Technical University (AKTU)** Lucknow, U.P., India
Bachelor of Technology – CSE (AI & ML) ; CGPA – 8.39/10 Sept 2024 – Aug 2028
Affiliated College: Shri Ramswaroop Memorial College of Engineering and Management (SRMCEM)

EXPERIENCE

- CAMPUS TECHNICAL CLUBS & LEADERSHIP**
 - Grobots Club (Robotics Club)** *Autonomous Domain Lead* May 2025 – Present
Represented institution in national autonomous robotics competitions; mentored 150+ members in PCB design, hardware timer isolation, PID tuning, and 3D body fabrication.
 - Computer Society of India (CSI) – SRMCEM Chapter** *Technical Lead* Apr 2025 – Aug 2026
Directed technical execution for 5+ chapter contests and bootcamps, mentoring 300+ students in software development and competitive programming.

PROJECTS

- HIGH-SPEED AUTONOMOUS ROBOTICS SUITE** *Embedded C/C++*
Multi-generation Autonomous Line-Follower & Maze-Solving Hardware Fleet
 - FireFly & BlueSky:** Developed baseline PID autonomous bots; optimized IR sampling & removed serial debug bottlenecks to boost speed by ~55% (110 to 170 PWM).
 - S6 (Impeller Suction Bot):** Engineered high-speed LFR with 1000 RPM motors and 8520 coreless motor suction fan for downforce at 255 PWM.
 - Line Tracer & Spirit:** Designed custom 16-channel IR array bot for sub-millimeter tracking; built lightweight *Spirit* bot with coreless drive-train.
 - Maze Solver:** Maze-solving bot with 3-loop averaged ADC sampling to eliminate sensor noise at high speeds.
 - Competition Results:** 2nd & 4th (MazeX, IIT BHU), 4th & 5th (LineQuest, IIT Guwahati), 1st & 3rd (BotFiesta, MUIT), 2nd (RoboRumble, CSJMU), 2nd & 4th (Technika, HBTU), 1st Runner-up (Utkarsh, BBDU).
 - Tech. Stack used** – Arduino Nano, PID Control, ADC Sensor Calibration, Custom PCB (16-ch IR), Impeller Fans, Embedded C/C++
- RISHIKA – VOICE-POWERED ROBOTICS TEACHING ASSISTANT** [Live Demo](#) [GitHub](#) 2025
 - Voice Mentorship & Diagnostics:** Architected low-latency (<800ms) WebSockets voice pipeline using LiveKit Agents, Deepgram STT, Gemini LLM, and Murf TTS; integrated SQLite context engine for Socratic hardware troubleshooting.
 - Tech. Stack used** – Next.js, LiveKit Agents, Deepgram STT, Gemini LLM, Murf TTS, SQLite, WebSockets
- UTILITY CHROME EXTENSIONS** [YT Shorts Blocker](#) [YT Ghoul](#) 2025
 - Browser Automation & DOM Filtering:** Engineered lightweight Manifest V3 Chrome extensions to filter distraction UI elements, manipulate DOM trees, and customize video playback workflows.
 - Tech. Stack used** – JavaScript (ES6+), Chrome Extension APIs (Manifest V3), HTML5/CSS3, DOM Manipulation

ACHIEVEMENTS

- HACKATHONS & COMPETITIONS**
 - HackMatrix 2.0 (IIT Patna)** – Voice AI Healthcare Platform 1st Place (Winner)
 - BotFiesta 2025 (MUIT Lucknow)** – Autonomous Line Follower 1st & 3rd Rank
 - SynergiX Hacks 2025 (VIPS TC)** – Top 30 of 2,000+ teams National Finalist
 - Make4Lucknow (AKTU & ERA)** – Farm2Market Agritech Platform Finalist
 - CodeX 2025 (SRMU)** – Python CP Contest 1st Runner-Up
 - Code Charades 3.0 (GFG)** – Offline WebDev Best UI/UX & 3rd Runner-up
- PROBLEM SOLVING & COMPETITIVE PROGRAMMING**
 - LeetCode** [Rating 1399 \(Peak 1414\) **Codeforces** \[Rating 962 \\(Max 1044\\)\]\(#\)](#)
 - HackerRank** [5★ Problem Solving & Python](#) **CodeChef** [Rating 1117 \(Peak 1234\)](#)

TECHNICAL SKILLS

- Languages:** Python, C, Embedded C/C++ (Arduino), JavaScript/TypeScript, SQL
- Frameworks & Developer Tools:** Next.js, FastAPI, LiveKit Agents, Supabase, SQLite, Git, Google Cloud Platform
- Robotics & Systems:** PID Control, ADC Sensor Calibration, Motor Drivers, Hardware Timers